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RESEARCH

Subdistrict-level drought assessment of agricultural lands in Vidarbha, Maharashtra, India using remote sensing-based indices

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Abstract
Drought is a prolonged shortage of water due to below-normal rainfall, and when such water deficits occur during critical stages of crop growth, they lead to agricultural drought and potentially severe crop losses. In Vidarbha, where drought poses a significant threat to agriculture, this study aims to evaluate crop health using remote sensing-based indices. The spatiotemporal variations of the Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI) sourced from the Mahalanobis National Crop Forecast Centre (MNCFC), Department of Agriculture, Cooperation and Farmers Welfare, Government of India, were analyzed from 2006 to 2021 at the sub-district level. Drought severity was evaluated using the Vegetation Condition Index (VCI), derived from NDVI, across multiple years. Seasonal NDVI, NDWI, and VCI were analyzed for kharif and rabi seasons using aggregated monthly data. Trend analysis using linear regression and the Mann-Kendall test showed positive trends across most tehsils, especially during rabi, supported by rising gram and wheat yields in Chandrapur. During the rabi seasons of 2013-14, 2016-17, 2017-18, 2019-20, and 2020-21 showed relatively higher NDWI values, consistent with enhanced NDVI, underscoring the linkage between vegetation health and water availability. A strong, significant ($p < 0.01$) positive correlation between NDVI and NDWI in July and September observed. These findings highlight the importance of remote sensing-based indices for timely and accurate crop health monitoring to mitigate the impacts of agricultural drought.

Keywords NDVI · NDWI · VCI · Mann-Kendall · Vidarbha · Drought

1 Introduction

As the "vertebral column" of human life, agriculture undoubtedly has a major impact on the economy (Ennouri et al. 2019). Agricultural drought significantly impacts the economy of agrarian nations like India, where over 68% of the population relies on agriculture for their livelihood. Approximately 16% of the country's geographical area is prone to drought, affecting nearly 50 million people each year (Dutta et al. 2015). Irregular and uneven spatial and temporal distribution of monsoon rainfall often leads to drought across extensive regions of India. A significant portion of the population relies on agriculture for their livelihood, recurrent droughts and related climatic risks pose substantial challenges to the growth and sustainability of the agricultural sector, thereby affecting its crucial role in poverty alleviation. The central parts of Maharashtra, particularly the Deccan Plateau and Vidarbha region, are recognized as drought-prone hotspots, underscoring the high agricultural vulnerability and climatic sensitivity of these areas (Ganjir et al. 2024).

The Vidarbha region, known for its fertile black cotton soil and diverse crop cultivation, has faced persistent agricultural distress marked by low productivity,

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inadequate irrigation infrastructure, and high rates of farmer suicides over the past 15 years (Vidarbha Weeping 2016). Challenges such as weather variability, limited agricultural knowledge, and increasing population pressure further strain this agrarian society. Agricultural droughts, triggered by water shortages during critical crop growth stages, often lead to crop failure. In this context, remote sensing has proven to be an effective tool for monitoring drought and assessing crops, providing affordable, frequent, and extensive spatial and temporal data coverage. Compared to localized ground-based measurements, satellite-derived indicators such as, soil moisture, evapotranspiration, temperature, and vegetation condition facilitate early detection and prompt intervention, thereby greatly improving drought monitoring and management efforts (Gu et al. 2008; Satapathy et al. 2024).

The increasing frequency of droughts has heightened the need for effective monitoring and management strategies to mitigate their impacts on agriculture and ecosystems. Remote sensing technology has revolutionized agricultural management and monitoring by providing real-time spatial information on crop health, soil conditions, weather patterns, and land use, while also enabling the estimation of key variables such as precipitation, evapotranspiration, soil moisture, and vegetation, crucial for assessing drought intensity, duration, and extent (Patel et al. 2012; Su et al. 2016; Ashraf et al. 2023). Remote sensing-based indices, such as the Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), Vegetation Condition Index (VCI), and Vegetation Health Index (VHI) are widely recognized tools for monitoring vegetation health, surface water availability, and drought dynamics at regional and global scales (Sharara et al. 2022; Xie and Fan 2021; Satapathy et al. 2024).

Vegetation conditions in drought prone regions are closely linked with key meteorological variables such as rainfall, temperature, humidity, and surface moisture. Many studies show that vegetation indices respond strongly to changes in these climate factors, making them reliable indicators for assessing vegetation stress (Choi et al. 2013; Eswar et al. 2016; Palmate et al. 2017; Bari et al. 2021; Das et al. 2023; Almouctar et al. 2024). NDVI is a widely utilized indicator for assessing crop stress and disease by capturing changes in photosynthetic activity (Rouse et al. 1973). The Vegetation Condition Index (VCI), introduced by Kogan (1995), enhances NDVI analysis by normalizing current values against long-term historical records, thereby offering a more context-sensitive evaluation of vegetation health over time. This makes NDVI and VCI effective for monitoring drought conditions (Zuzulova and Vido 2018; Nicholson and Farrar 1994; Kogan 1995; Seiler et al. 2000; Wang et al. 2001; Anyamba et al. 2001; Ji and Peters 2003; Dutta et al. 2015). NDWI, developed by Gao (1996), is more sensitive to leaf water content and canopy wetness, providing critical insights into water stress in plants (Thomas et al. 2004; Chakraborty and Sehgal 2010; Shashikant et al. 2021). In recent years, researchers globally have increasingly employed these indices to study the impact of climatic change on agriculture and vegetation.

Numerous studies across India have employed remote sensing-based indices to monitor agricultural drought and assess vegetation health at both national (Kundu et al. 2016; Thanabalan et al. 2023) and regional levels, including Tamil Nadu (Prajesh et al. 2019), Maharashtra (Aswathi et al. 2018), Rajasthan (Dutta et al. 2015), Gujarat (Bhukya et al. 2023; Sur and Lunagaria 2020), Rayalaseema (Sandeep et al. 2021), and Bundelkhand (Danodia et al. 2021). The Mann-Kendall test (Mann 1945; Kendall 1975), a non-parametric method, widely applied in climatology and hydrology was employed alongside these indices to detect spatiotemporal trends in time series data. Though not predictive, it effectively identifies significant long-term changes in drought frequency and severity, offering valuable insights for drought assessment and early warning systems.

2 Materials and methods

2.1 Study area

Vidarbha, located in the eastern part of Maharashtra, India (Fig. 1), comprises the Nagpur and Amravati Divisions. Covering 31.6% of Maharashtra's total area and housing 21.3% of its population (Dandekar 2014), Vidarbha shares borders with Madhya Pradesh to the north, Chhattisgarh to the east, Telangana to the south, and the Marathwada and Khandesh regions of Maharashtra to the west. This central Indian region spans 97,950 km^2 and consists of 11 districts and 120 sub-districts (tehsils), positioned between latitudes 19.05° to 21.25° N and longitudes 75.59° to 79.11° E (Nair and Mirajkar 2021). Situated on the northern Deccan Plateau, Vidarbha is rich in black cotton soil, dense forests, and

Fig. 1 Study area map of Vidarbha region, Maharashtra, showing district and tehsil boundaries and location within India

abundant mineral resources. The Wainganga River is the region's largest, along with major tributaries such as the Wardha, Kanhan, and Painganga, all of which flow into the Godavari River in the south.

The region experiences a semi-arid climate and is heavily dependent on rainfall for agriculture. Temperature extremes ranging from highs of 47 °C in summer to lows of 12 °C in winter, with July and August receiving the most rainfall, and April receiving the least. The eastern part of Vidarbha receives an average annual rainfall of 1500 mm, while the western region gets about 700 mm, with 85–90% of the total rainfall occurring during the Southwest Monsoon from June to September (Nair and Mirajkar 2021). Predominantly a low-rainfall area, the region relies heavily on rain-fed farming, covering 89% of cultivated land where cotton, jowar (Barley), and pulses are the main crops (Deshmukh and Lunge, 2013).

In Vidarbha, Kharif crops cultivated during the monsoon season (June-October) include cotton, rice, soybean, tur (pigeon pea), and maize, while Rabi crops (October-March) comprise wheat, jowar, gram, linseed, and mustard. The eastern subregion, covering Nagpur, Bhandara, Chandrapur, Gondia, and Gadchiroli is recognized as the "rice belt," whereas the western districts of Wardha, Amravati, Akola, Yavatmal, Washim, and Buldhana form the "cotton belt". Irrigation coverage remains limited, with only 47.4% of the potential utilized compared to 76.4% in Western Maharashtra. Although Vidarbha has a larger paddy cultivation area (7.06 lakh ha), its average productivity (780 kg ha^{-1}) is markedly

lower than that of Western Maharashtra (1850 kg ha^{-1}) (Roy 2014; Joshi and Lepse 2017). The region has experienced severe drought episodes, notably during El Niño years (2002, 2004, 2009), with vulnerability increasing since 2012, leading to drought occurrences almost every alternate year (Gupta and Jain 2020; Nair and Mirajkar 2022; Dohale et al. 2024).

2.2 Data

The data employed in this study was sourced from the Mahalanobis National Crop Forecast Centre (MNCFC), under the Department of Agriculture, Cooperation and Farmers Welfare, Government of India, New Delhi. The datasets, including the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Water Index (NDWI), were obtained through the MNCFC's National Agricultural Drought Assessment and Monitoring System (NADAMS) portal (https://ncfc.gov.in/nadams.html). MNCFC derived these indices by averaging values specifically over agricultural areas, so they only represent cultivated land and not natural vegetation. Furthermore, the Vegetation Condition Index (VCI) used in this study was calculated directly from the NDVI dataset.

2.2.1 NDVI

The NDVI data available on the MNCFC portal is generated under the NADAMS project. The satellite and sensors used to derive NDVI include RESOURCE SAT 2 - AwiFS composite data and TERRA MODIS VI product (MOD13Q1) 16-day composite data at 250 m spatial assessment. The main purpose for the development of this data set is drought assessment. The NDVI is derived using the formula:

$$\mathrm{NDVI} = \frac{(\mathrm{NIR} - \mathrm{RED})}{(\mathrm{NIR} + \mathrm{RED})}$$

Where NIR is near-infrared in the infrared channel and RED is the red band in the visible channel.

Monthly NDVI data for the sub-district level (Tehsil/Taluka level) from 2006 to 07 to 2020-21 was downloaded for the current study.

2.2.2 NDWI

NDWI is also generated under the NADAMS project using TERRA MODIS VI PRODUCT (MOD13Q1) 16-day composite data at 250-meter spatial resolution for the drought assessment. Monthly NDWI is available for download at the sub-district level. The data were downloaded for the period 2006-07 to 2020-21 for the Vidarbha region. NDWI is derived using the formula:

$$\mathrm{NDWI} = \frac{(\mathrm{NIR} - \mathrm{SWIR})}{(\mathrm{NIR} + \mathrm{SWIR})}$$

Where NIR and SWIR are the reflected radiation in Near Infrared and Shortwave Infrared channels.

2.2.3 VCI

Vegetation Condition Index (VCI) (Kogan 1995) was computed using the NDVI data. VCI compares the current NDVI to the range of values observed in the same period in previous years. This index has been used by researchers across the world for detecting the onset of drought and measuring its intensity, duration, and impact. It is one of the methods for normalizing NDVI values. VCI is computed using the formula:

$$\mathrm{VCI_J} = \frac{\mathrm{NDVI_I} - \mathrm{NDVI_{min}}}{\mathrm{NDVI_{max}} - \mathrm{NDVI_{min}}} \times 100$$

Where $\mathrm{NDVI_{max}}$ and $\mathrm{NDVI_{min}}$ are calculated from the long-term record for that month/week/season and J is the index of the current month (week/season).

2.3 Methods used

2.3.1 Constructing seasonal series

The data obtained were on a monthly scale and were used to analyze variations during the kharif and rabi seasons. However, the seasonal datasets had to be constructed from the monthly data. For the kharif season, the series was generated by averaging the NDVI, NDWI, and VCI values for the months of June, July, August, and September at the sub-district level. A similar approach was applied for the rabi season, using the data from October, November, December, and January to create the seasonal series.

2.3.2 Representation of Spatial variation

The sub-district (tehsil) level shapefile of Vidarbha was sourced from the Survey of India's portal. This shapefile played a crucial role in accurately depicting the spatial

and temporal variations of NDVI, NDWI, and VCI across the region. These indices were employed to examine and visualize changes on monthly, seasonal, and annual scales, enabling a thorough evaluation of vegetation health, water availability, and drought conditions over time. This mapping approach offered important insights into the spatiotemporal dynamics of agricultural areas in Vidarbha.

2.3.3 Trend analysis

Trends, in NDVI and NDWI for all the tehsils, were calculated using the non-parametric Mann-Kendall test (Mann 1945; Kendall 1955) and linear regression.

The Mann-Kendall statistic S is calculated based on the number of positive and negative differences between data points at different time steps. For a time-series of n data points, the test examines all possible pairs of data values and computes the difference between each pair.

Let the time series data be represented as $x_1, x_2, \ldots, x_n$, where x_i is the value at time t_i. For each pair of data points (x_i, x_j), where $j > i$, the difference $x_j - x_i$ is calculated. The sign function sgn is then applied to the difference:

$$\operatorname{sgn}(x_j - x_i) = \begin{cases} +1 & if\ x_j > x_i \\ 0 & if\ x_j = x_i \\ -1 & if\ x_j < x_i \end{cases}$$

The test statistic S is the sum of the signs of all possible differences between the data pairs:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \operatorname{sgn}(x_j - x_i)$$

A positive S indicates an increasing trend, while a negative S signifies a decreasing trend and If S is close to zero, it indicates no significant trend.

To assess the significance of S, the variance Var(S) is calculated. If there are no tied (equal) data points, the variance is computed as:

$$\operatorname{Var}(S) = \frac{n(n-1)(2n+5)}{18}$$

where n is the number of data points in the time series.

If there are ties (i.e., repeated values in the dataset), the variance calculation must account for them.

$$\operatorname{Var}(S) = \frac{1}{18}\left[n(n-1)(2n+5) - \sum_{p=1}^{g} t_p(t_p-1)(2t_p+5)\right]$$

Where, g is the number of tied groups and t_p is the number of observations in the p^{th} group.

Once S and Var(S) are computed, the test statistic *Zmk* is calculated to determine the significance of the trend. The Z-statistic standardizes the S value by dividing it by its standard deviation σs = $\sqrt{\operatorname{Var}(S)}$

$$Z_{MK} = \begin{cases} \frac{S-1}{\sqrt{\operatorname{Var}(S)}} & if \quad S > 0 \\ 0 & if \quad S = 0 \\ \frac{S+1}{\sqrt{\operatorname{Var}(S)}} & if \quad S < 0 \end{cases}$$

A positive Z value indicates an increasing trend, while a negative Z value indicates a decreasing trend. The significance of the trend was evaluated at the 90% confidence level ($p < 0.10$), 95% confidence level ($p < 0.05$) and 99% confidence level ($p < 0.01$). Then the value of Z is compared against a critical value from the standard normal distribution to determine whether the trend is statistically significant.

2.3.4 Linear regression and correlation

To examine trends in the Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI), the data were plotted using Microsoft Excel. Monthly values of both indices were aggregated over time to identify seasonal and annual patterns. Additionally, a scatter plot was created to assess the correlation between NDVI and NDWI. The Pearson correlation coefficient was calculated from the plotted data to quantify the strength and direction of the linear relationship between the two indices. This helped in understanding how vegetation health (NDVI) is related to water content (NDWI) across the study region.

3 Results and discussion

3.1 Seasonal changes in NDVI

3.1.1 Kharif season

Agriculture in Vidarbha largely relies on rainfall, with many crops being rainfed. Figure 2 shows the seasonal NDVI progression over Vidarbha closely reflects the rainfed kharif cropping pattern. Low NDVI values in June (0.1–0.3) correspond to bare soil and initial sowing after monsoon onset, while July (0.3–0.5) indicates early vegetative growth of cotton, soybean, paddy, and pulses. By August, NDVI rises

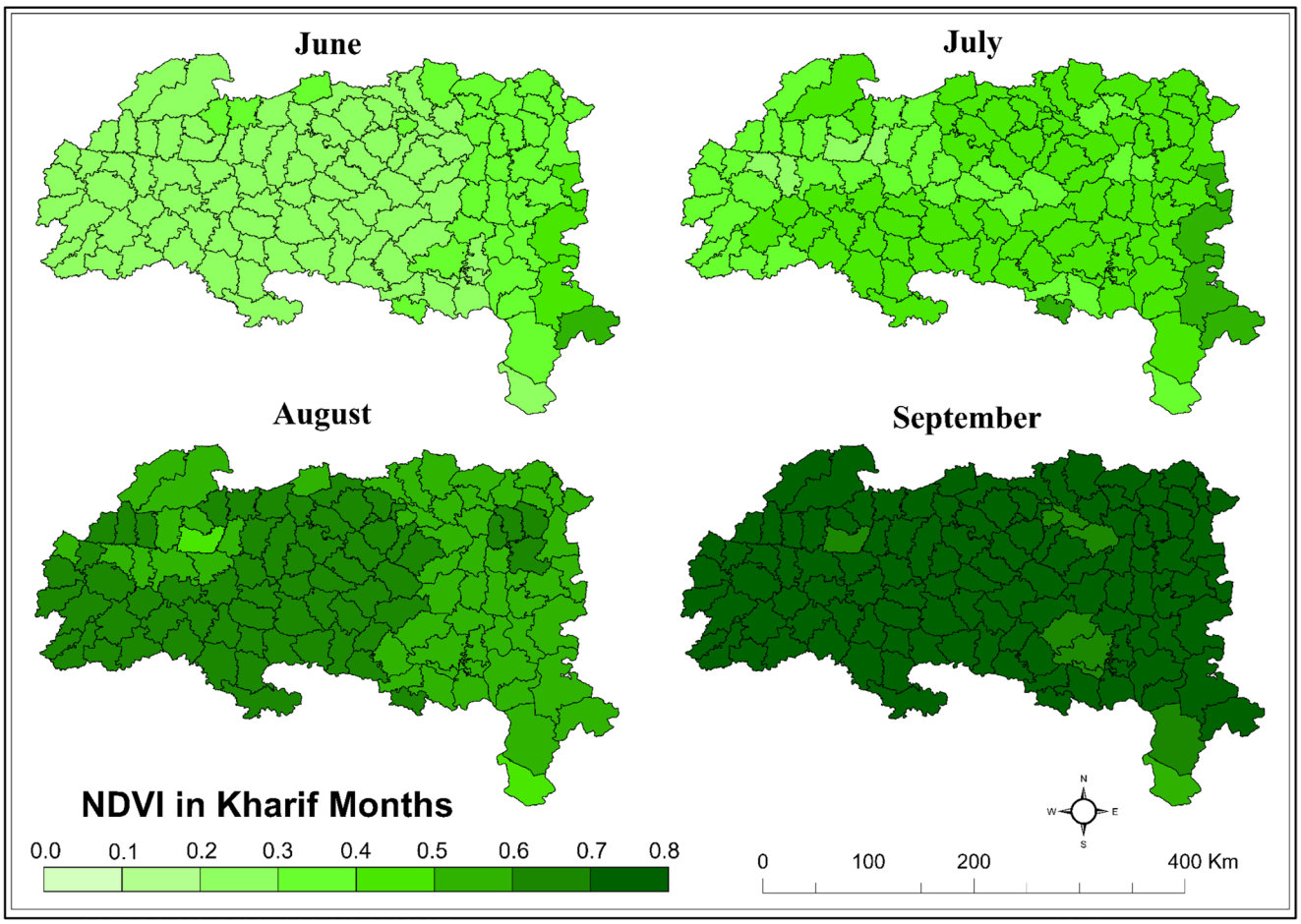

Fig. 2 Spatial variation of average NDVI across tehsils in Vidarbha during the kharif season (June-September), reflecting crop growth stages under rainfed agriculture

to 0.5–0.7, marking peak vegetative cover, and reaches its maximum in September (0.7–0.9) with dense canopy. This spatiotemporal pattern highlights the strong dependence of kharif agriculture on monsoon rainfall and reveals localized variability in crop growth.

Figure 3 shows the relationship between monthly average rainfall and NDVI over Vidarbha. Rainfall peaks in July, whereas NDVI values continue to rise and reach their maximum in September, indicating a delayed vegetation response to rainfall. The delayed response of vegetation to rainfall is evident in the NDVI and rainfall trends due to crop growth and soil moisture accumulation (Chandrasekar et al. 2006; Vyas et al. 2015). NDVI remained relatively high through October despite declining rainfall, reflecting the continued use of residual moisture. The decline in NDVI during November-January coincided with the harvest of kharif crops and the dry season. These findings emphasize that vegetation indices respond not only to immediate rainfall but also to cumulative moisture availability, highlighting their relevance for agricultural drought monitoring.

The inter-annual variations in NDVI during the kharif season were studied across Vidarbha and represented in Fig. 4. It was observed that the higher NDVI values in years such as 2006-07, 2007-08, 2010-11, 2011-12, 2015-16, 2017-18 and 2020-21 indicate favorable rainfall and healthy crop growth, while lower values in 2009-10, 2013-14, 2014-15, 2018-19 and 2019-20 suggest drought stress and reduced vegetation growth. Spatial heterogeneity is evident, with southern and eastern parts often showing higher greenness compared to central and western regions. Overall, NDVI effectively captures the

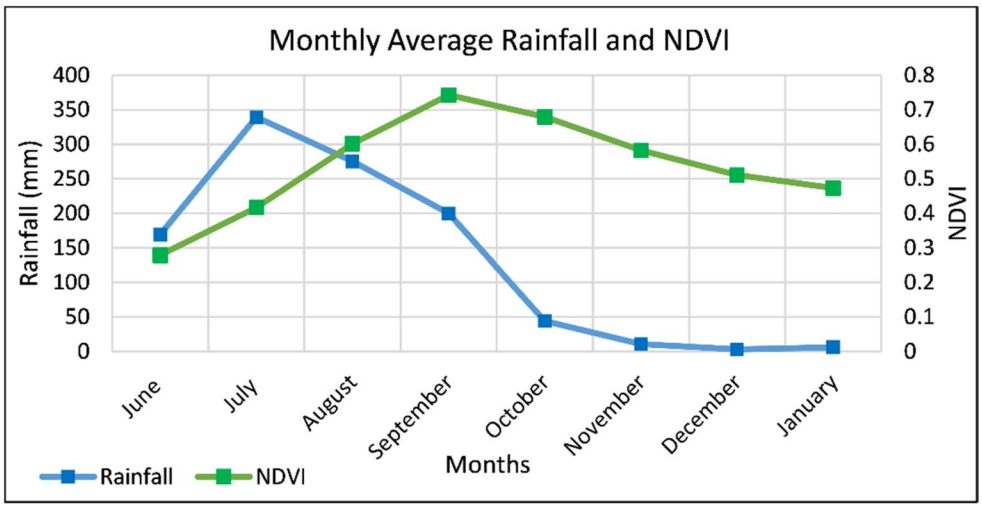

Fig. 3 Monthly average rainfall and NDVI over Vidarbha, representing 15-year averages (2006–2021) from June to January

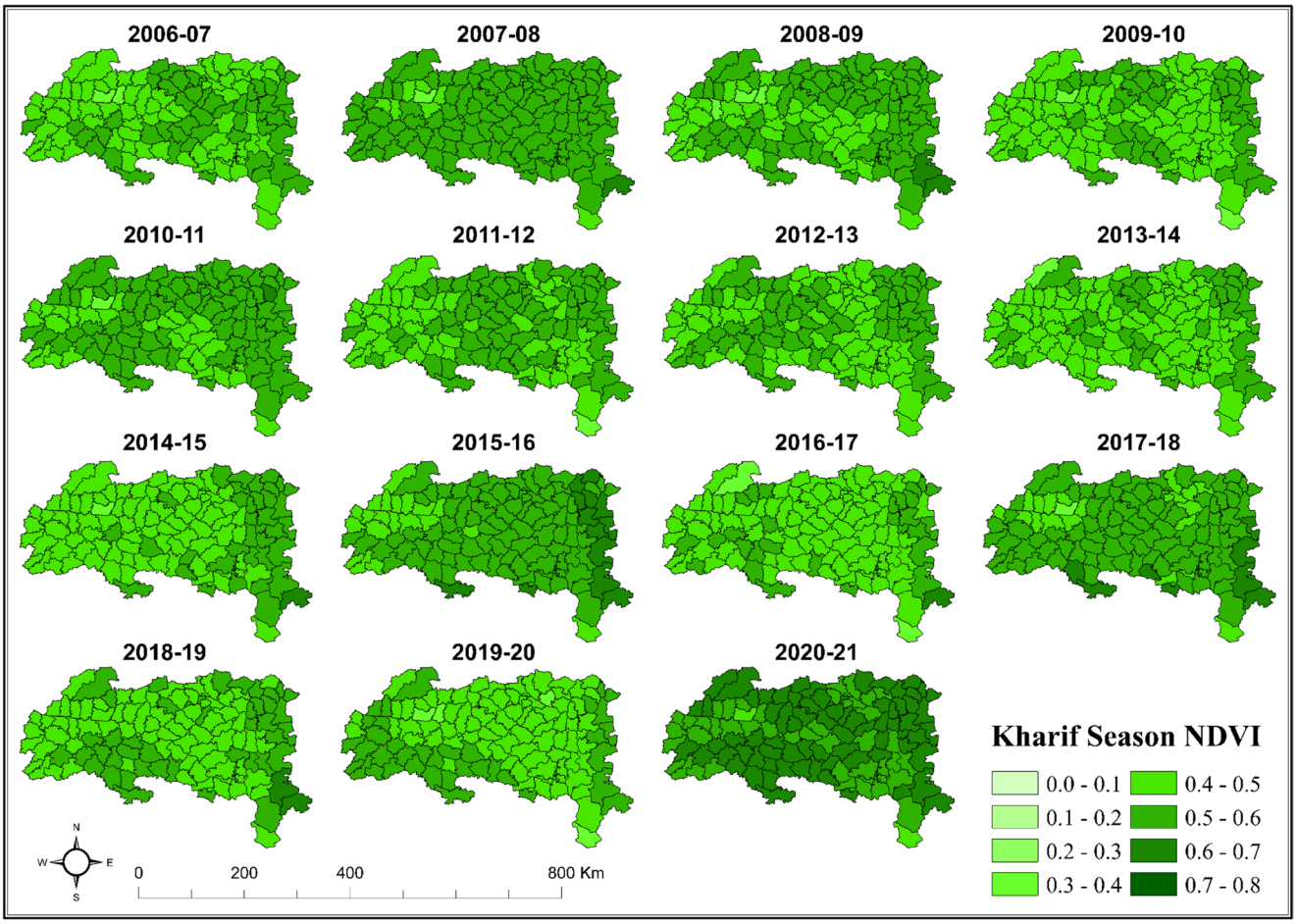

Fig. 4 Spatiotemporal variations in average NDVI during the kharif season across the study area (2006–2020)

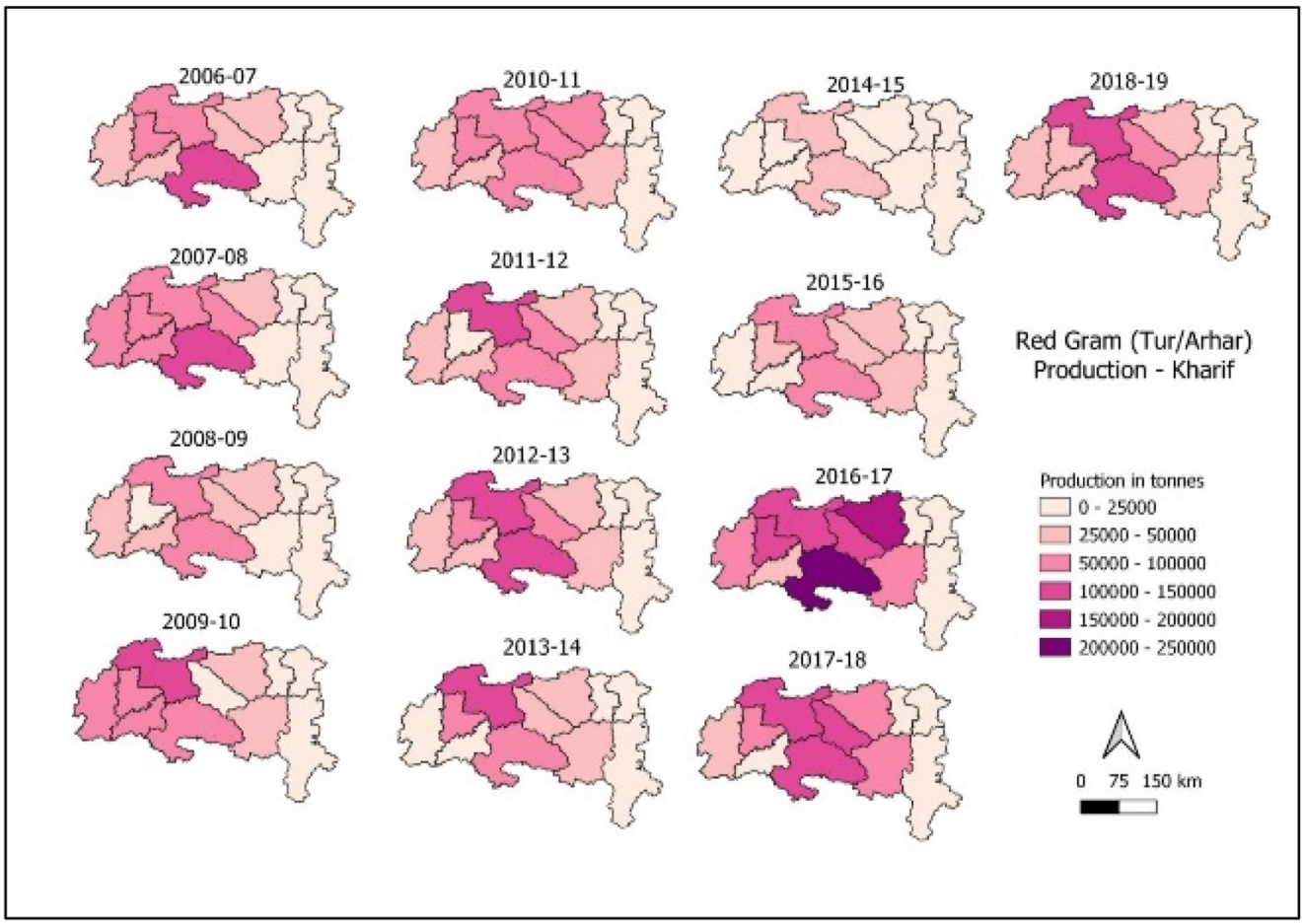

Fig. 5 Spatiotemporal distribution of red gram production in kharif across districts of Vidarbha (2006–2019)

sensitivity of vegetation to monsoon variability, making it a useful indicator of agricultural performance and drought impacts.

Since tehsil-level crop production data was unavailable, district-level statistics were obtained from https://data.gov.in/catalog/district-wise-season-wise-crop-production-statistics. Despite inconsistencies in data continuity across crops and districts, red gram (tur/arhar) production as a kharif pulse crop records were relatively complete, making them suitable for analysis. Figure 5 represents district-wise red gram production across Vidarbha (2006–07 to 2018–19), revealing clear spatial and temporal patterns influenced by agro-climatic factors and cropping systems. Western districts like Amravati, Akola, Yavatmal, and Washim consistently show higher yields due to black soil, moderate rainfall, and prevalent cotton-tur intercropping. In contrast, eastern districts such as Bhandara, Gondia, Gadchiroli, and Chandrapur exhibit lower production, shaped by forested landscapes, higher precipitation, and rice-centric agriculture. While direct correlation with NDVI is complex, elevated NDVI values in 2006-07, 2007-08, 2012-13, 2016-17, 2017-18 and 2018-19 align with higher Tur output in central and western districts, indicating favorable vegetation health and crop performance during the seasons. This integrated analysis underscores NDVI's utility for crop performance and climate responsiveness in semi-arid agricultural landscapes.

3.1.2 Rabi season

During the monsoon season, NDVI values over Vidarbha peak in September, reaching ~ 0.9, which reflects the vigorous vegetative growth of kharif crops supported by abundant rainfall. However, Fig. 6 show the harvesting of major crops such as soybean, paddy, and cotton

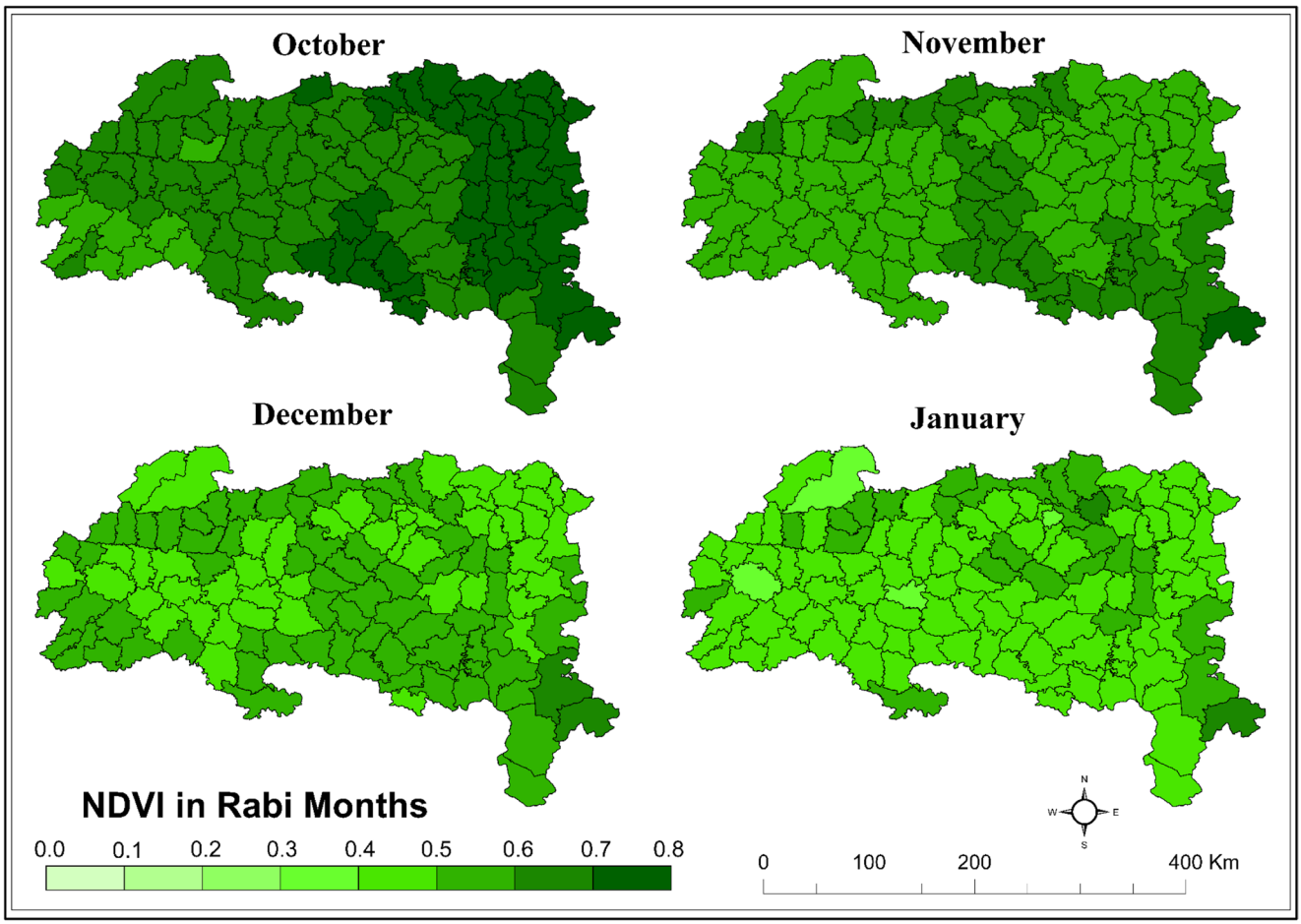

Fig. 6 Spatial variation of average NDVI across tehsils in Vidarbha during the rabi season (October-January)

beginning in October, NDVI values decline to ~0.6–0.7, particularly in the central and western parts of the region. From November, NDVI continues to decrease as harvesting is largely completed, and most croplands are left fallow. In December, NDVI values further drop to ~0.3–0.5, showing limited vegetation cover, which highlights the absence of widespread rabi cropping due to the predominantly rainfed nature of agriculture in the region. Only irrigated tehsils, maintain higher NDVI (~0.5–0.7), representing wheat, gram, or other irrigated rabi crops. By January, NDVI values fall further to ~0.2–0.4, indicating bare soil or sparse vegetation across most of the region. Overall, the seasonal NDVI trend clearly captures the agricultural cycle in Vidarbha, with peak greenness during kharif, sharp decline after harvest, and minimal rabi cultivation except in irrigated areas.

Rabi season NDVI in Fig. 7 shows pronounced inter-annual variability with a consistent east-west gradient. Higher NDVI is observed in 2016-17, 2017-18, and 2019-20, while 2007-08 and 2008-09 show distinctly lower values. Spatially, eastern Vidarbha (Gondia, Bhandara, Chandrapur, Gadchiroli) consistently records NDVI values > 0.5, reflecting irrigated rabi cropping, whereas central-western districts (Wardha, Akola, Washim, Amravati) mostly remain below 0.5, indicating poor or negligible rabi coverage under rainfed conditions. Years with higher NDVI correspond to rabi crops reaching maturity during October-December and benefitting from residual soil moisture and favorable post-monsoon rainfall. Conversely, years with NDVI < 0.5 highlight the predominance of fallow lands or moisture stress. Overall, NDVI values effectively capture the variability in rabi cropping extent and productivity across years, governed by both irrigation availability and antecedent monsoon rainfall.

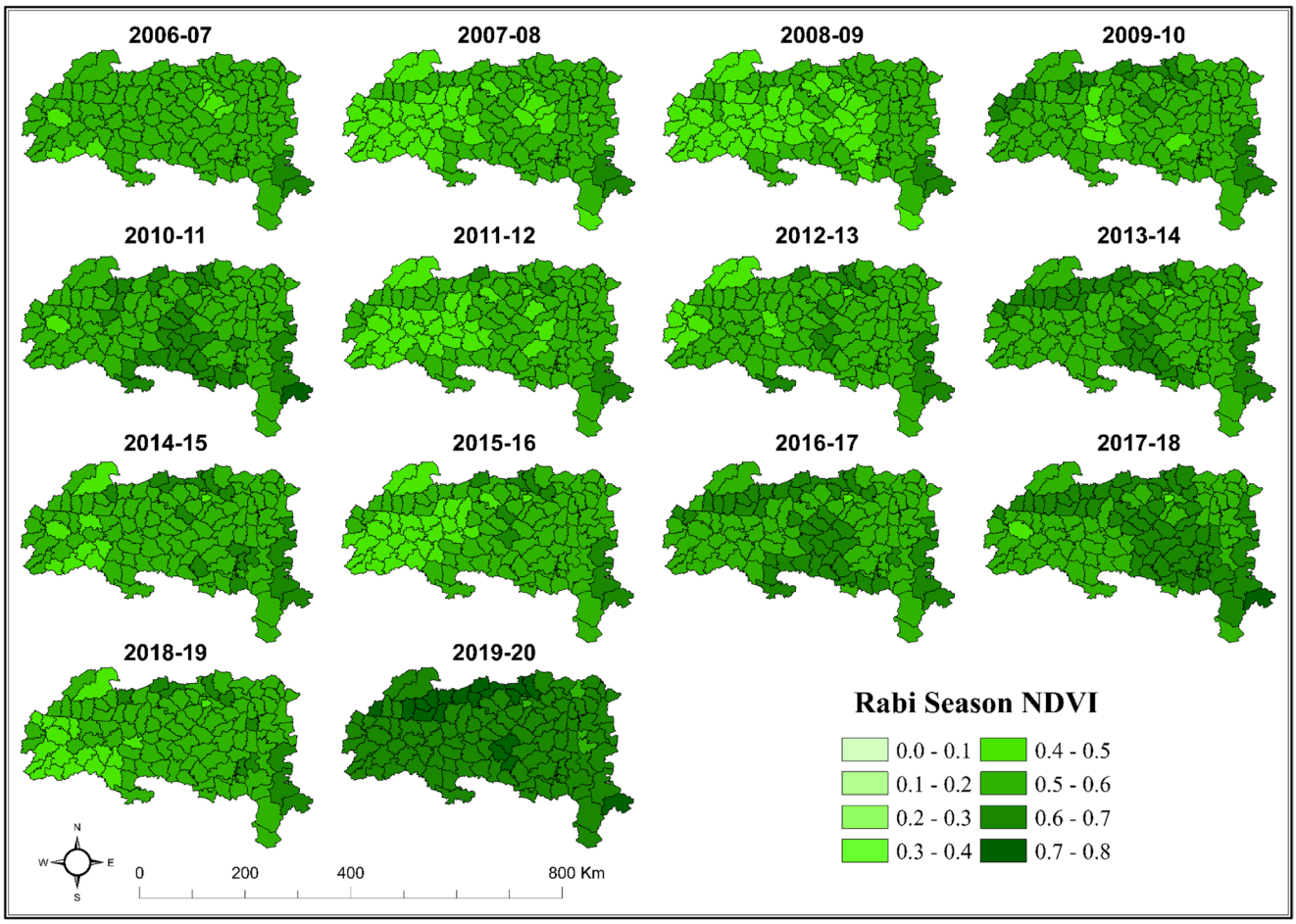

Fig. 7 Spatiotemporal variations in average NDVI during the rabi season across the study area (2006–2020)

3.2 Seasonal changes in NDWI

3.2.1 Kharif season

The progression of NDWI during the kharif season was analyzed and shown in Fig. 8. NDWI values in Vidarbha progressively increased from June (0–0.2.2) to August (0.4–0.7), reflecting crop establishment, peak vegetative growth, and abundant soil, canopy moisture supported by monsoon rainfall. June, being the first month of the southwest monsoon, experiences drier conditions compared to the later months, as reflected in the maps with June having the lowest NDWI values and September the highest. As the monsoon progresses, more moisture becomes available to crops, which they absorb for growth. Although crops generally begin to lose moisture as they approach maturity in September, NDWI remained high (0.5–0.7), suggesting that prolonged monsoon activity and the presence of long duration crops such as cotton and paddy sustained highest canopy water content.

Water content in vegetation fluctuates seasonally and annually, depending on monsoon rainfall. The annual NDWI maps for the kharif season (Fig. 9) closely resemble the corresponding NDVI maps for the same period, as shown in the seasonal NDVI changes. Years with favorable NDVI conditions reflecting good NDWI values, likely due to crop water content is closely linked to crop health. For instance, years with higher NDWI (2007-08, 2011-12, 2013-14, 2016-17, 2017-18, 2020-21) correspond to enhanced NDVI, indicating favorable rainfall and crop growth, while drought years (2008-09, 2009-10, 2014-15, 2018-19, 2019-20) showed concurrent declines in both indices, reflecting rainfall deficit, water stress, and reduced vegetation.

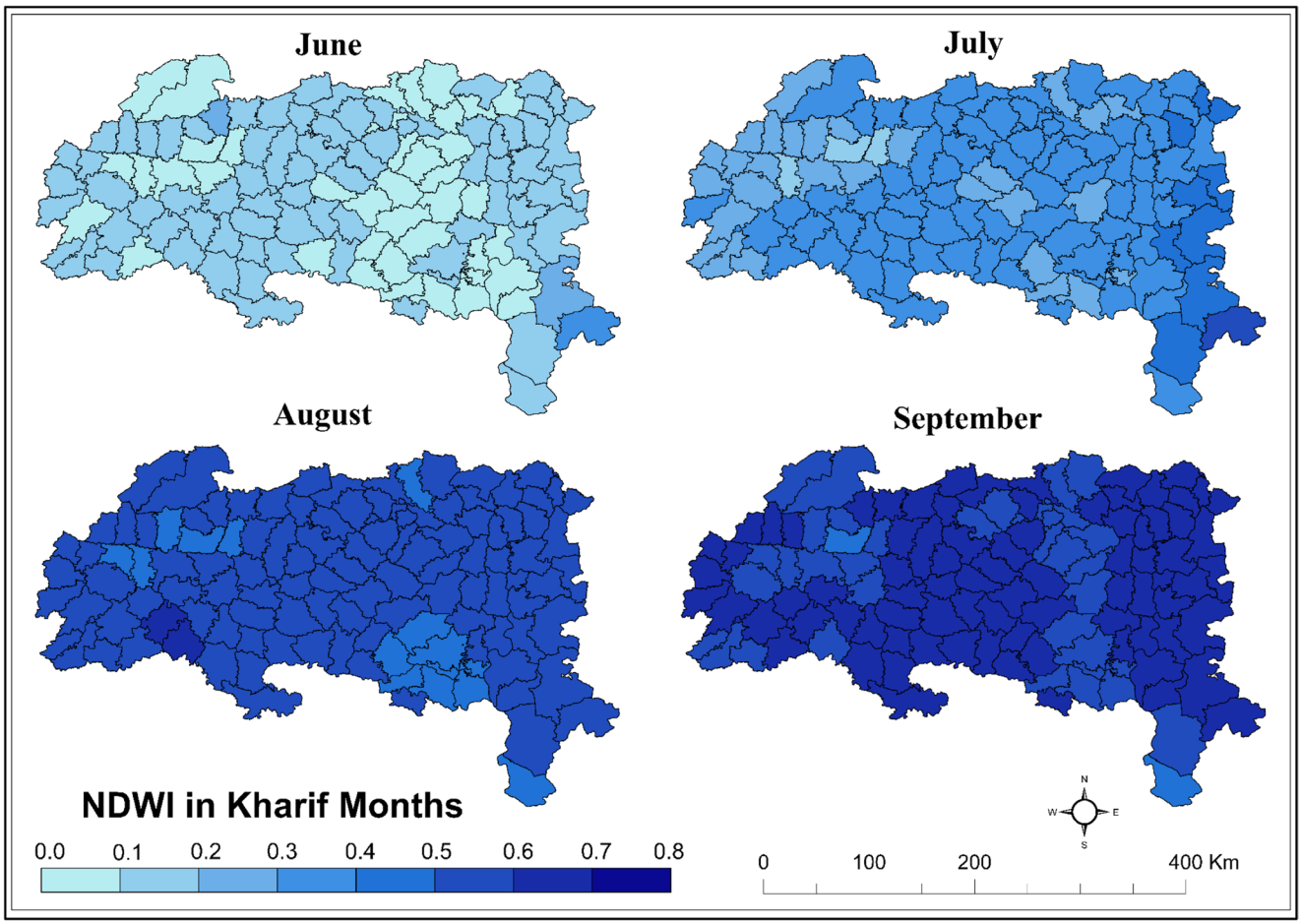

Fig. 8 Monthly NDWI variation in Vidarbha during June-September, showing seasonal progression in vegetation and soil moisture

3.2.2 Rabi season

Following the withdrawal of the southwest monsoon by the end of September, rainfall decreases sharply and arid conditions begin to dominate in Vidarbha. Despite this, Fig. 10 shows NDWI values in October remain moderately high (0.3–0.7), particularly in the eastern tehsils of Bhandara, Gondia, Chandrapur, and Gadchiroli, where values reach 0.6–0.7 due to residual soil moisture and standing late-stage crops. By November, NDWI values decline to 0.2–0.4, reflecting progressive drying of soils and reduced vegetation water content. The decreasing trend continues into December and January, with NDWI falling below 0.3 in December and reaching 0.2 by January. This seasonal decline is primarily associated with the harvesting of kharif crops and the predominance of fallow land during the rabi season, as irrigation facilities remain limited across much of the region.

The Rabi season NDWI maps (2006-07 to 2020-21) Fig. 11 shows marked interannual variability in soil moisture and vegetation water content across Vidarbha. Years such as 2008-09, 2009-10, 2014-15, 2015-16, and 2018-19 recorded predominantly low NDWI values (0.2–0.4), reflecting limited water availability, whereas 2010-11,2013-14, 2016-17, 2017-18, and 2019-20 exhibited higher NDWI (0.5–0.7), indicating favorable crop water conditions. These NDWI values showed a strong spatial and temporal correspondence with NDVI patterns across Vidarbha. Spatially, eastern districts consistently showed higher NDWI than western areas, highlighting the influence of rainfall distribution and irrigation.

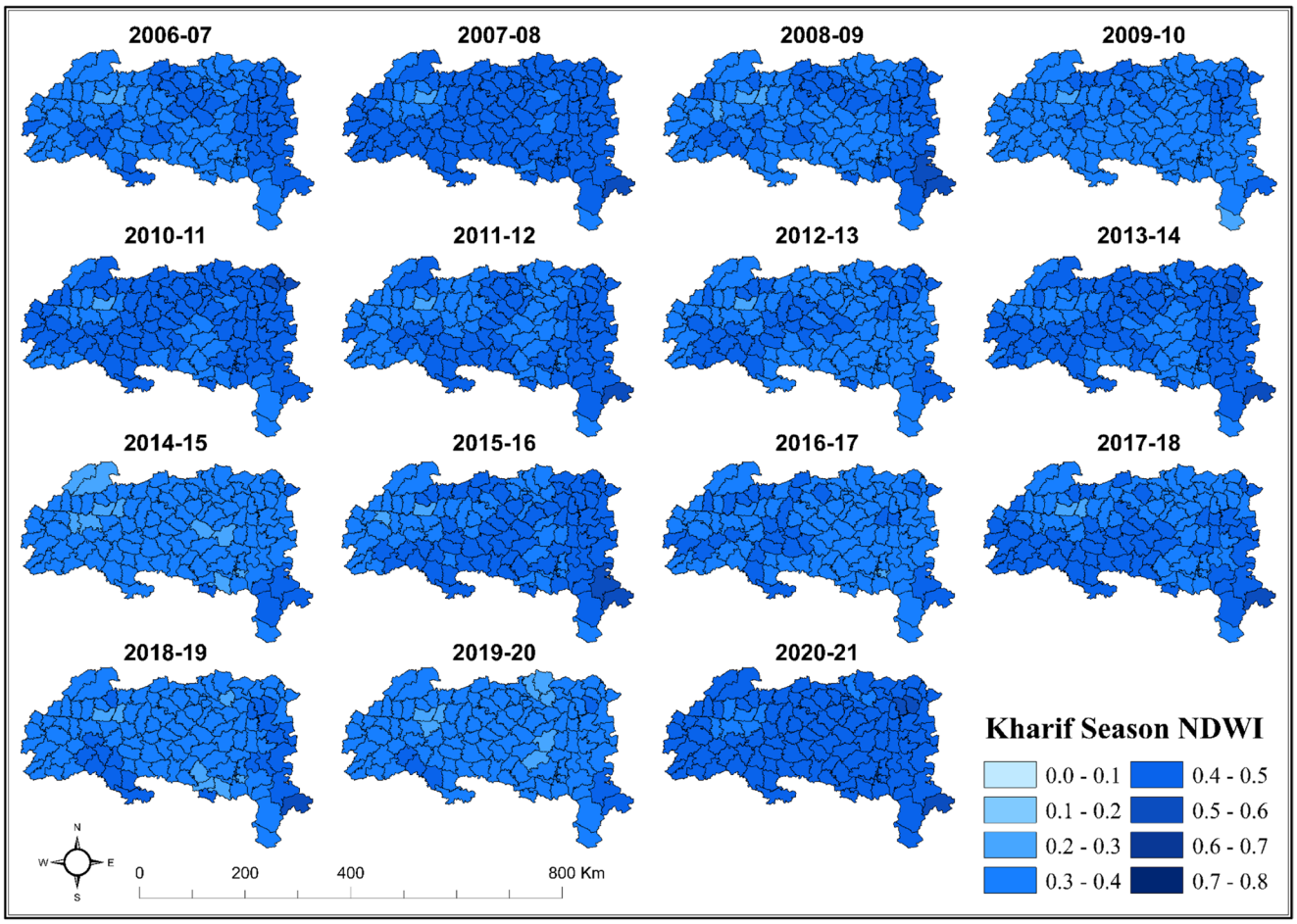

Fig. 9 Spatiotemporal variations in average NDWI during the kharif season across the study area (2006–2020)

3.3 Seasonal changes in VCI

The Vegetation Condition Index (VCI) was derived from NDVI values and multiplied by 100 to express in percentage. Seasonal maps of the VCI were made for each year from 2006 to 07 to 2020-21. VCI values from 0 to 10% indicate extreme drought, 10% to 20% severe drought, 20% to 30% moderate drought, 30% to 40% mild drought, above 40% normal conditions, and above 80% are considered normal.

The Kharif season VCI in Fig. 12 shows pronounced spatiotemporal variability in drought severity. Years such as 2009-10, 2013-14, 2014-15, 2016-17, and 2019-20 exhibit large areas with VCI below 40%, reflecting mild to extreme drought conditions linked to deficient monsoon rainfall and crop stress. In contrast, years like 2007-08, 2015-16, 2017-18, and 2020-21 record widespread VCI above 40%, with 2020-21 showing values exceeding 80% in most regions, indicating favorable rainfall and normal vegetation growth. Transitional years such as 2006–07, 2008–09, 2010-11, 2011–12, 2012–13, and 2018–19 reflect moderate vegetation conditions, with VCI between 40 and 60%. Overall, the VCI effectively captures the alternation of drought and favorable years, strongly governed by monsoon variability.

The Rabi VCI maps (Fig. 13) show pronounced drought conditions in 2006-07, 2007-08, 2008-09, 2011-12, 2012-13, 2015-16, and 2018-19, with VCI < 40% indicating severe vegetation stress, particularly in 2008-09 and 2011-12. Relatively better conditions occurred in 2009-10, 2010-11, 2013-14, 2016-17, and 2017-18 (VCI 40 − 0%), while 2019-20 recorded widespread favorable vegetation with VCI > 80% reflecting excellent vegetation health supported by favorable rainfall and soil moisture. Spatially, drought stress was more severe in western and central Vidarbha, whereas eastern parts

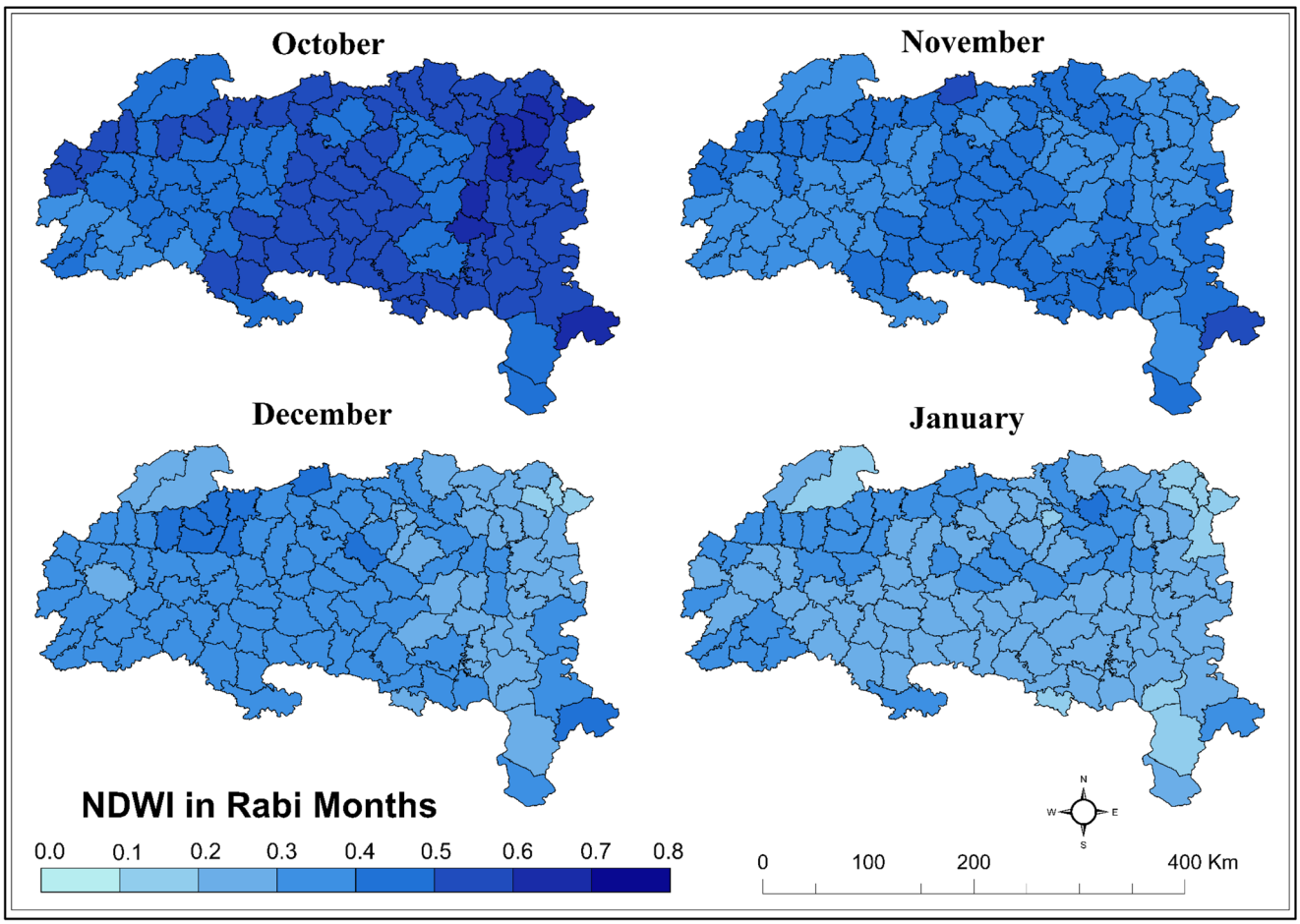

Fig. 10 Monthly NDWI variation in Vidarbha during October-January, showing seasonal decline in vegetation and soil moisture

often retained comparatively better vegetation cover. This highlights the strong interannual variability of Rabi season droughts in Vidarbha.

3.4 Trend analysis

3.4.1 Trends in NDVI

To analyze the NDVI trends over the study period across all 120 tehsils in Vidarbha, both linear regression and the non-parametric Mann-Kendall test (Z-statistic) were employed. The linear regression results showed that almost all tehsils exhibited positive NDVI trends in rabi season and partial tehsils in kharif season, suggesting improvements in crop health and more favorable growing conditions. This indicates that crops are healthier now than in the past.

The Mann-Kendall trend analysis confirms a steady increase in NDVI values over the past 15 years, particularly during the rabi season. Nearly all tehsils show an upward NDVI trend, with only Nagpur tehsil showing a slight decrease, though not statistically significant. In total, 41 tehsils display a statistically significant increase at the 99% confidence level, 28 tehsils at the 95% confidence level, and 11 tehsils at the 90% confidence level. The remaining 39 tehsils also show an upward trend, though not statistically significant, indicating an overall improvement in crop health during the rabi season across the study period (Figs. 14 and 15).

For the kharif season, the trends are more mixed. Of the 120 tehsils, some show increasing trends, while others show decreases. Mauda and Ashti tehsils exhibit significant decreases, with Mauda at the 95% confidence level and Ashti at the 90% confidence level. Additionally, 41 tehsils show non-significant decreases, and 12 show no trend at all. Conversely, four tehsils display a statistically significant increase at the 95% confidence level, three at the 90% confidence level, and 58 tehsils show increasing trends that are not statistically

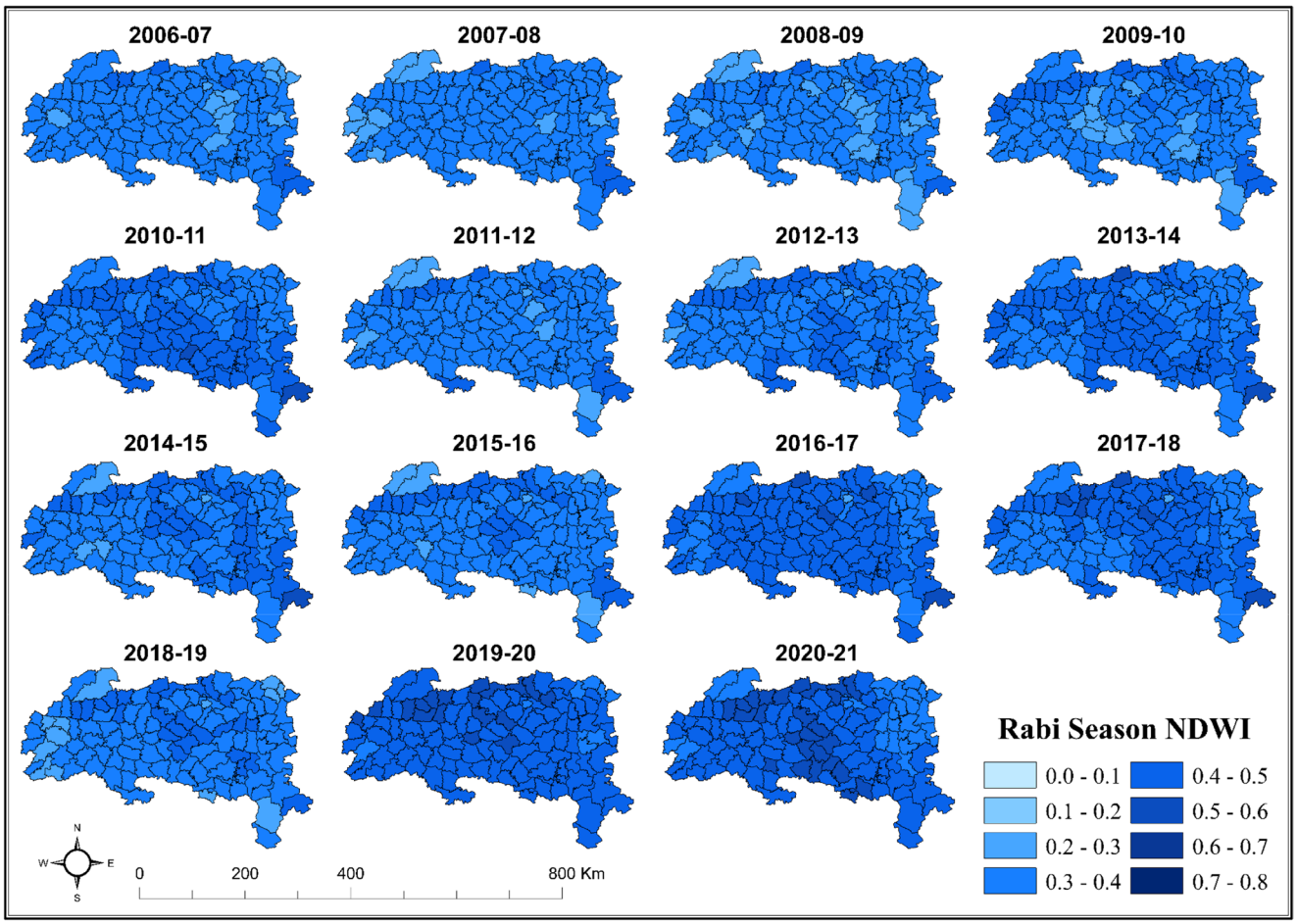

Fig. 11 Spatiotemporal variations in average NDWI during the rabi season across the study area (2006–2020)

significant. This mixed trend suggests more variability in crop health improvements during the kharif season (Figs. 14 and 15).

3.4.2 Trends in NDWI

Linear regression analysis showed that nearly all tehsils in Vidarbha experienced positive NDWI trends in the Rabi season, with only some tehsils showing this trend during the Kharif season. This suggests improved crop health and growing conditions over time, indicating healthier vegetation than in previous years. The Mann-Kendall test corroborated this, showing a consistent NDWI increase over the past 15 years, especially in the Rabi season. All 120 tehsils displayed an upward trend in NDWI: 49 tehsils had a statistically significant increase at the 99% confidence level, 37 at 95%, and 18 at 90%, while the remaining 16 showed positive but non-significant trends, collectively suggesting better crop health in Rabi (Figs. 14 and 16).

In the Kharif season, trends were more varied. Some tehsils showed increases, while others showed declines. Mauda tehsil had a significant decrease at the 99% confidence level, with five other tehsils showing significant decreases at the 90% level. Additionally, 69 tehsils showed non-significant decreases, and 12 showed no trend, while 33 tehsils displayed non-significant increases (Figs. 14 and 16). These mixed Kharif season trends reflect diverse environmental and crop growth conditions across the region.

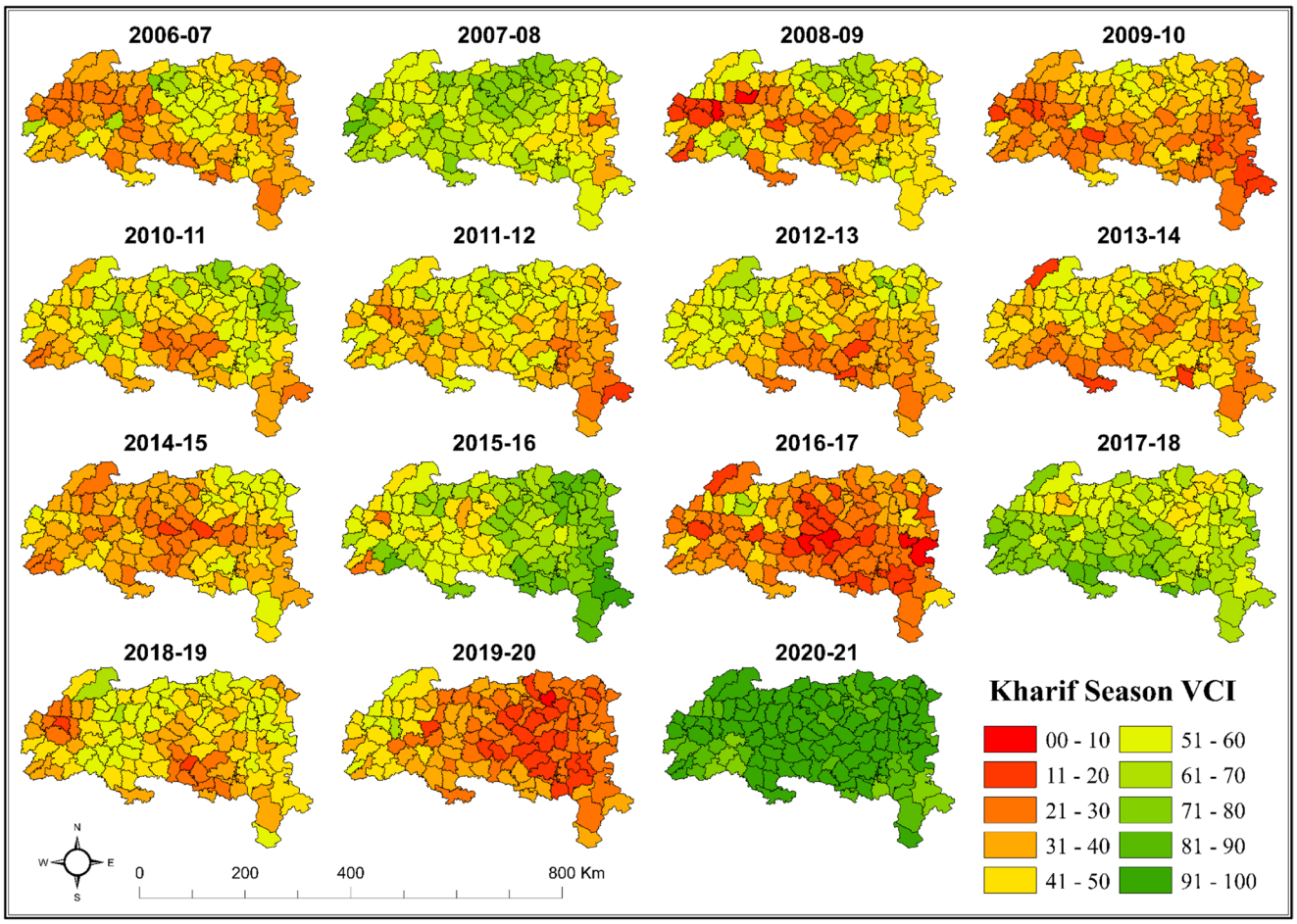

Fig. 12 Spatiotemporal variability of VCI (%) during kharif season in Vidarbha (2006 to 2020)

3.4.3 Trends in VCI

To examine the trends in VCI over the study period across all 120 tehsils in Vidarbha the non-parametric Mann-Kendall test (Z-statistic) were employed. Figure highlights trends in the Vegetation Condition Index (VCI) for the Rabi and Kharif seasons in Vidarbha over the study period. During the Rabi season, most tehsils show favorable VCI trends, indicating improved vegetation conditions. Specifically, 11 tehsils exhibited a statistically significant increasing trend at the 99% confidence level, 35 tehsils at the 95% level, and 9 tehsils at the 90% level. An additional 58 tehsils showed a positive but non-significant trend, suggesting overall enhancements in crop health and growing conditions in Rabi across Vidarbha. Only a few tehsils demonstrated decreasing trends, with 1 tehsil showing a significant decrease at the 95% confidence level and 4 tehsils displaying a non-significant decline (Figs. 14 and 17).

In contrast, the Kharif season presents more mixed trends in VCI. Here, 10 tehsils showed a significant decreasing trend at the 95% confidence level, and 3 tehsils at the 90% level, indicating some areas with deteriorating vegetation conditions. Additionally, 58 tehsils displayed a non-significant decrease, while 8 tehsils showed no trend at all. Conversely, 41 tehsils exhibited an upward trend in VCI, though these increases were not statistically significant (Figs. 14 and 17), suggesting minor improvements in vegetation health during the Kharif season in parts of Vidarbha.

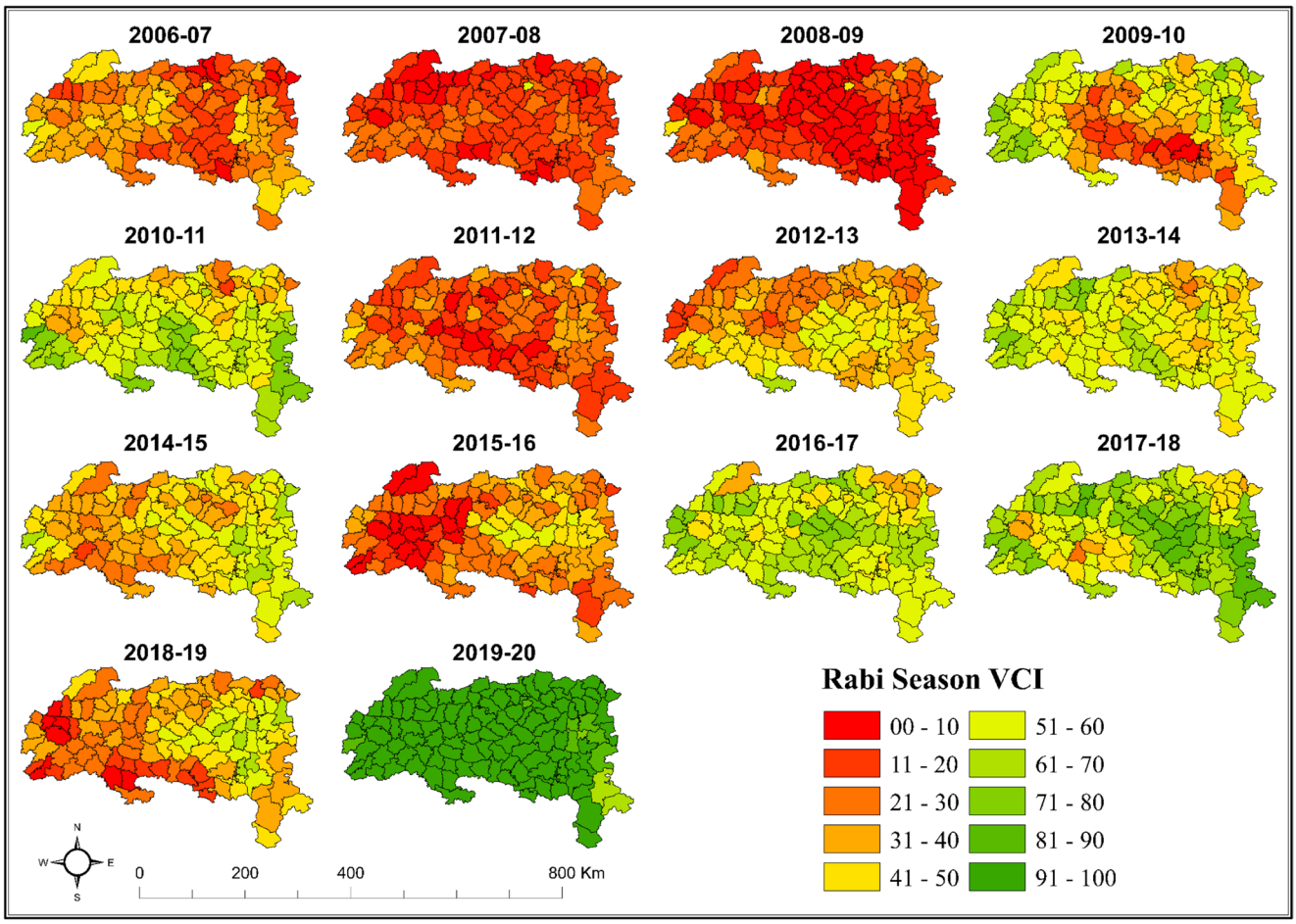

Fig. 13 Spatiotemporal variability of VCI (%) during rabi season in Vidarbha (2006 to 2019)

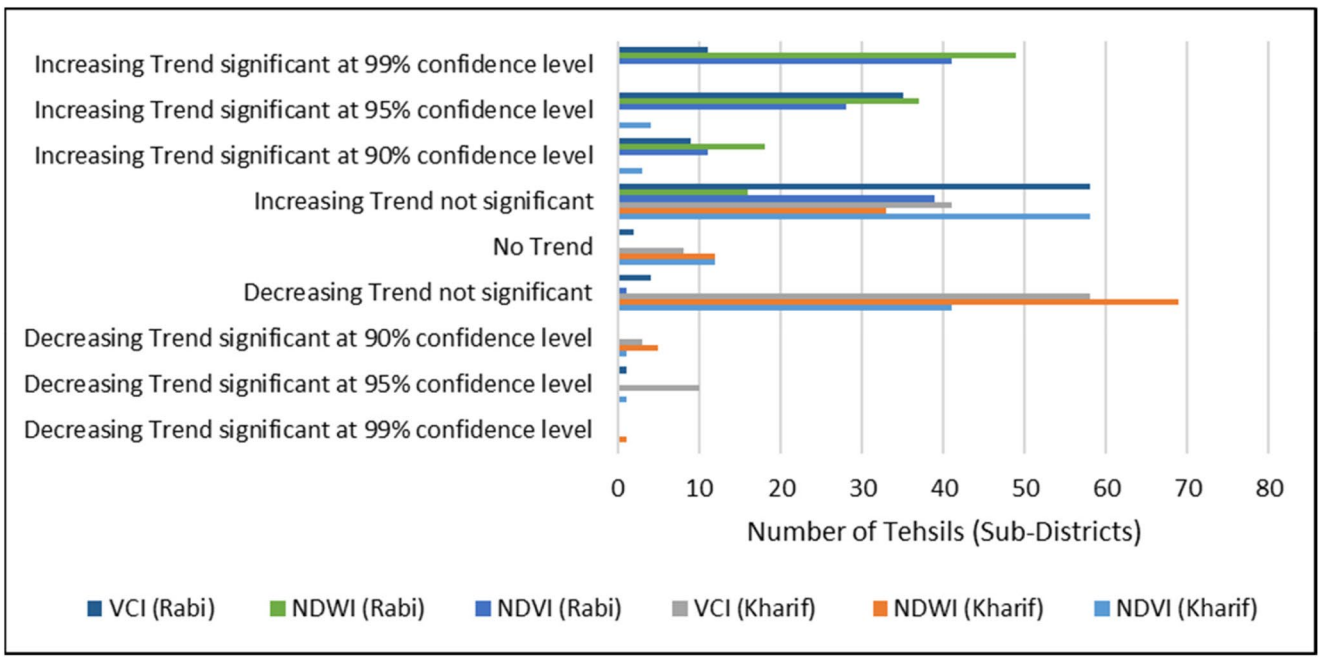

Fig. 14 Interpretation of the Z statistic for the trends in NDVI, NDWI, and VCI during kharif and rabi seasons

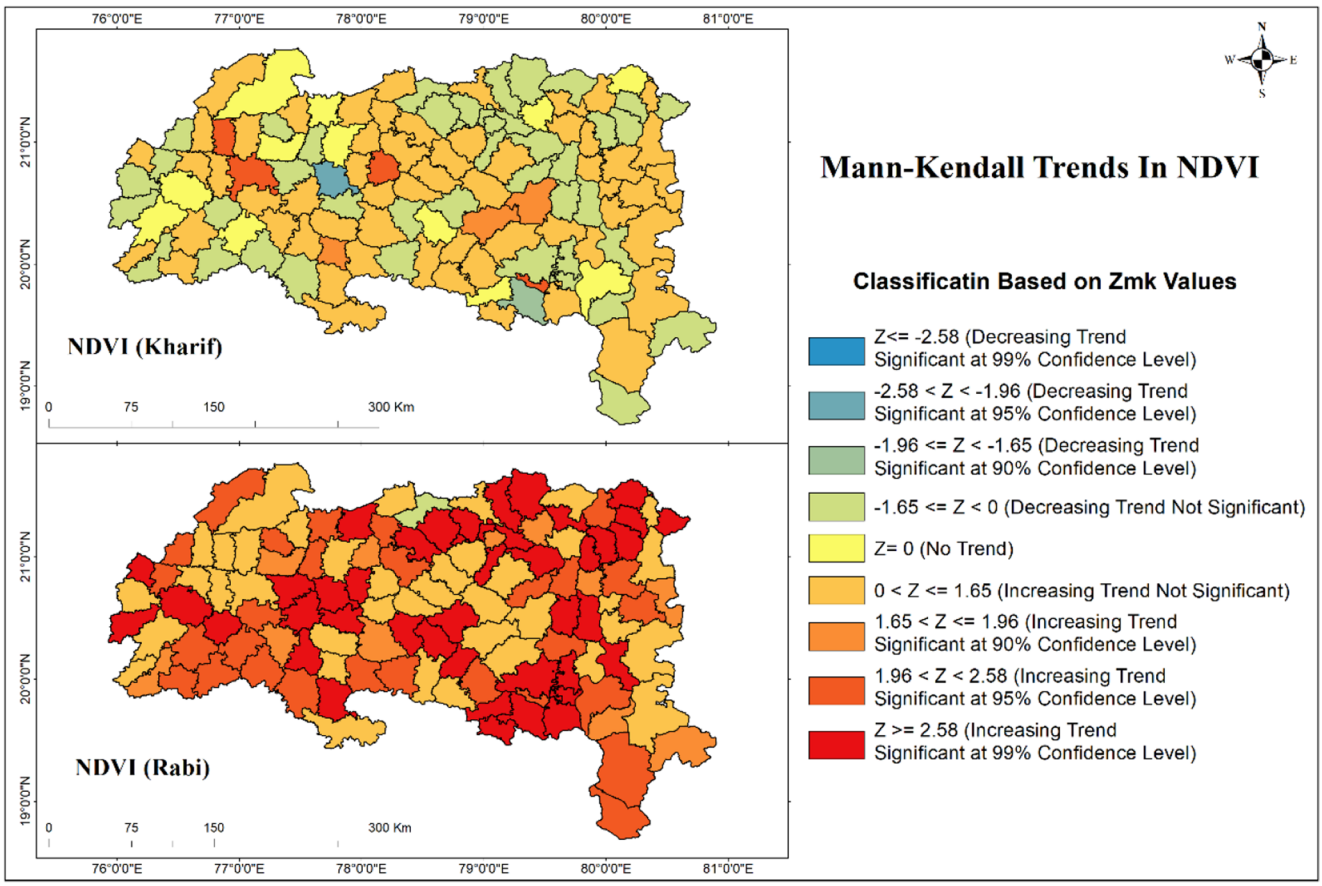

Fig. 15 Mann Kendall trend analysis of seasonal NDVI during kharif and rabi seasons over Vidarbha

3.5 Relationship between NDVI and NDWI

Understanding the relationship between NDVI and NDWI is crucial for comprehending how changes in one index affect the changes in other. For this purpose, the values of NDVI and NDWI for each kharif month of 15 years were plotted and a bivariate model was fitted to the scatterplot obtained. These plots for June, July, August, and September are depicted in Fig. 18. In all the graphs it is observed that there is a strong positive relationship between these two indices. An increase in NDWI leads to an increase in NDVI. For June, the squared correlation coefficient is 0.27 which increases to 0.89 during July. The association between NDWI and NDVI is stronger in July. In June, conditions are mostly arid and crops are not yet established, so no clear association is observed. But in July the agricultural fields have crops that are feeding on the soil moisture increasing plant water content and leading to the good health of crops. During August the correlation value decreases but the direction of the relationship is the same. The correlation in September again increases to 0.85 indicating a strong positive relationship between NDWI and NDVI. The analysis reveals a positive and significant relationship between NDWI and NDVI across all kharif months at the 0.01 level (99% confidence).

4 Conclusion

The main objective of the study was to study geographical variation in NDVI and NDWI over Vidarbha and to comprehend drought conditions using VCI. Results show

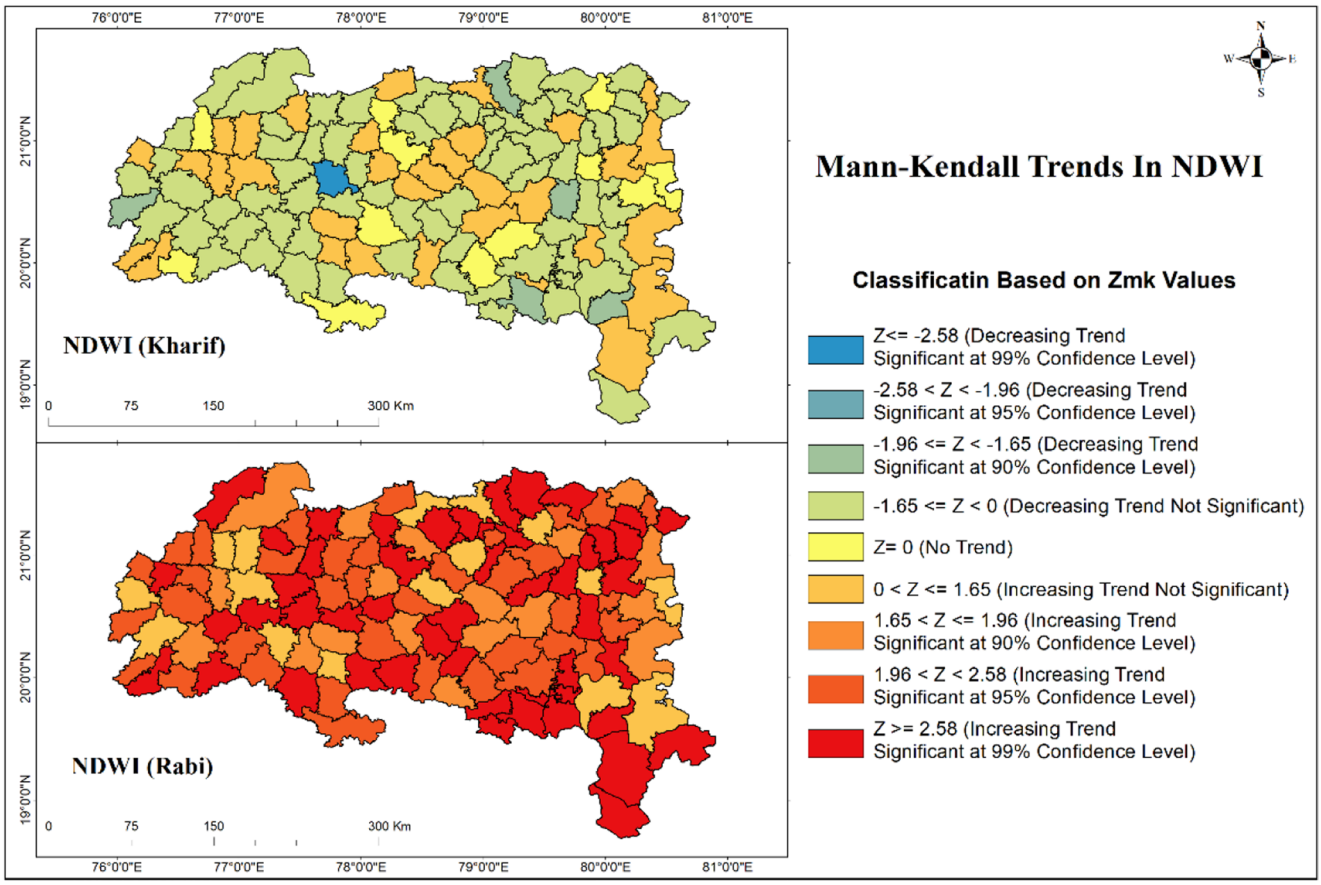

Fig. 16 Mann Kendall trend analysis of seasonal NDWI during kharif and rabi seasons over Vidarbha

that NDVI and NDWI values increase from June to October after which they subside except for some pockets in Vidarbha. Vegetation has a lagged response to rainfall as NDVI peaks in September and rainfall peaks in July. In the kharif season, NDVI peaks with the onset of the monsoon, accompanied by an increase in NDWI. VCI has revealed that the region experienced severe drought conditions during the rabi season of 2006−07, 2007-08, 2008-09, 2011-12, 2012-13, 2015-16, and 2018-19. The Seasonal comparison over the past 15 years, indicates that kharif generally supports healthier vegetation cover compared to rabi, highlighting the strong influence of seasonal rainfall on vegetation dynamics.

The rabi seasons of 2013-14, 2016-17, 2017-18, 2019-20, and 2020-21 showed relatively higher NDWI values, consistent with enhanced NDVI, underscoring

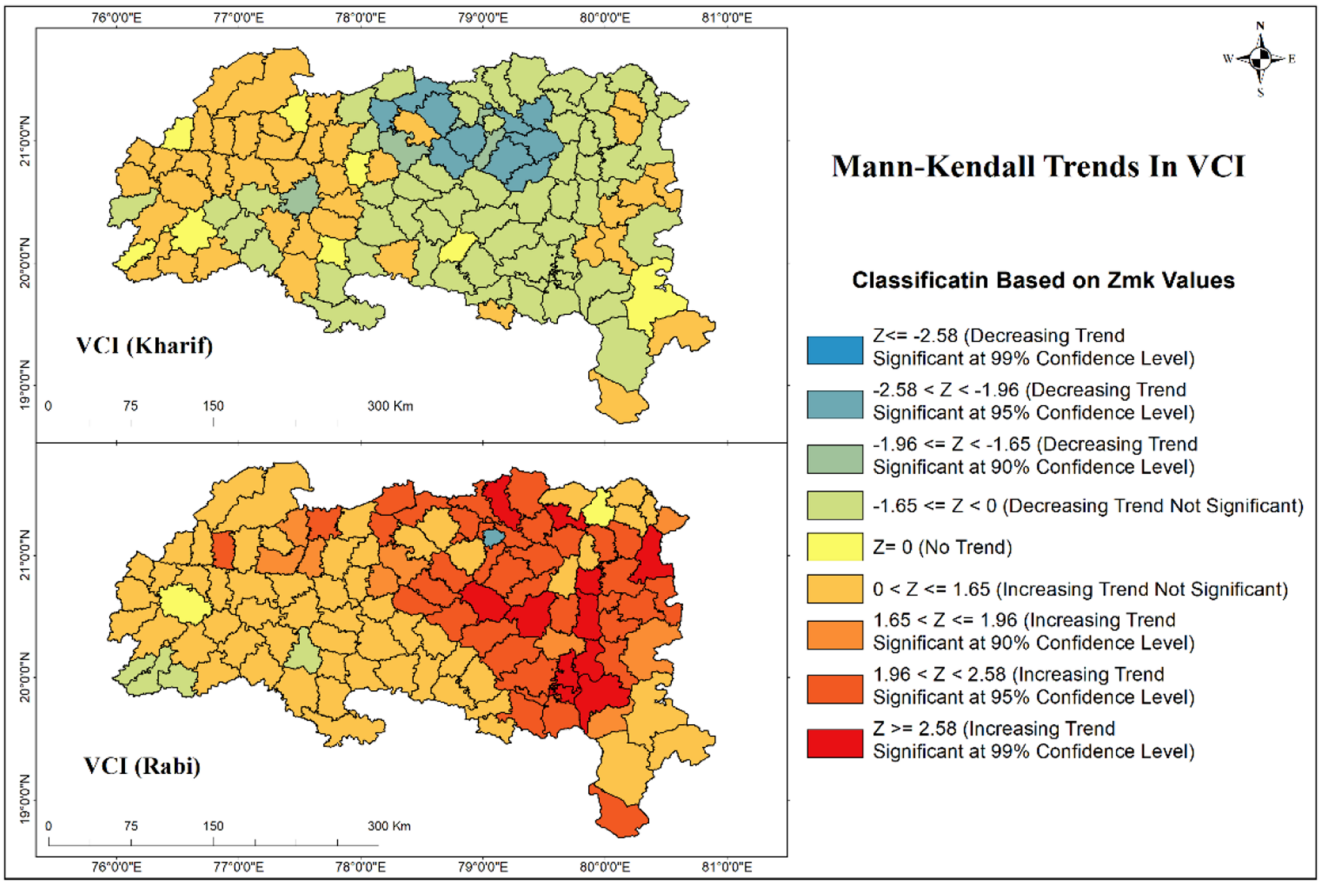

Fig. 17 Mann Kendall trend analysis of seasonal VCI during kharif and rabi seasons over Vidarbha

the linkage between vegetation health and water availability. A significant positive association (at 0.01 level) between NDVI and NDWI during the kharif season further confirms that monsoon failure directly translates into moisture stress and drought impacts. Overall, rabi exhibits stronger positive trends in vegetation health and moisture content compared to kharif, reflecting cumulative effects of seasonal rainfall and favorable growing conditions.

Vidarbha, traditionally recognized as a water-stressed region, has shown an increasing trend in NDVI and NDWI over time, suggesting gradual improvement in agricultural conditions. However, recurring droughts continue to pose serious risks to farmers' livelihoods. Strengthening water conservation practices and improving moisture

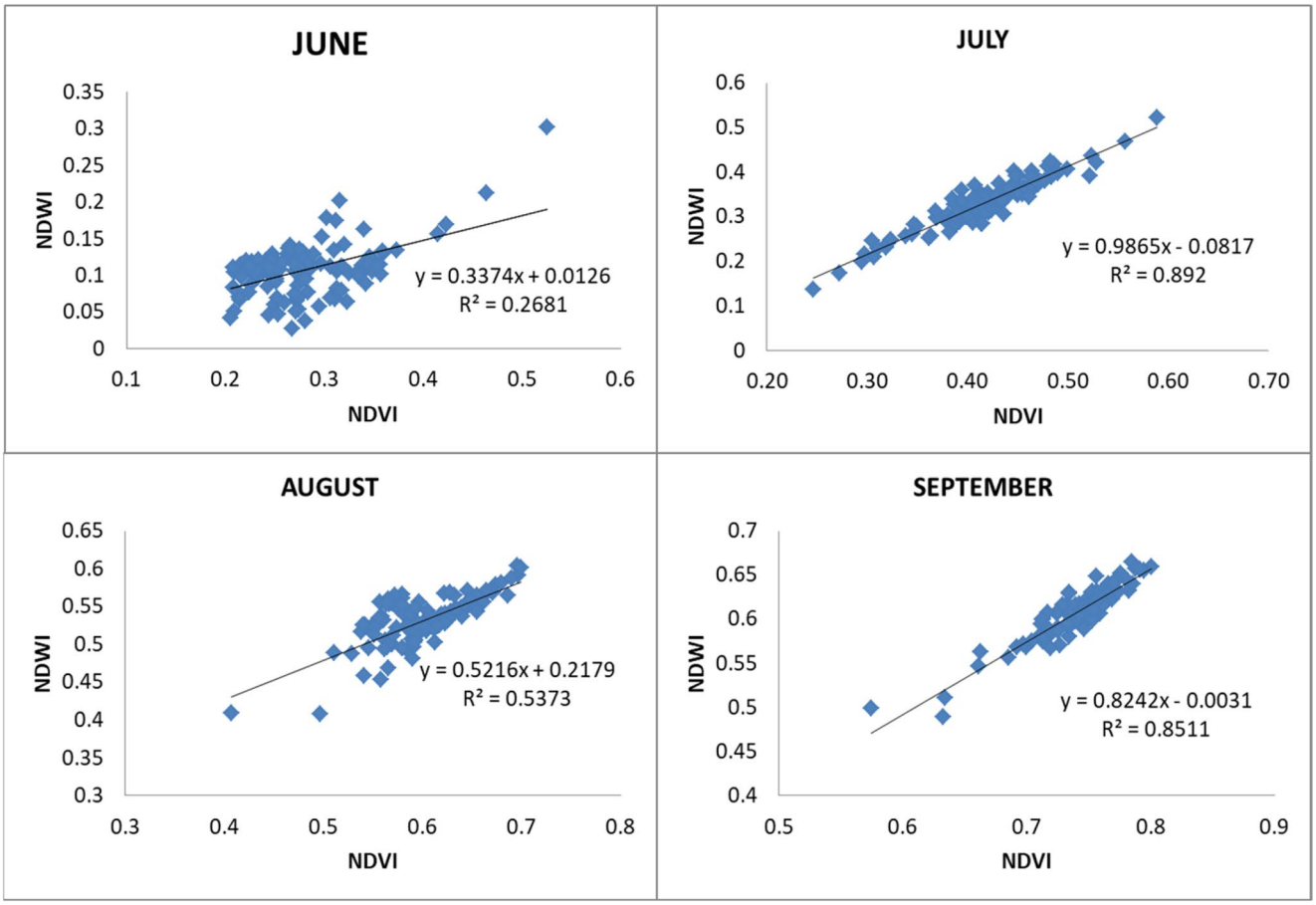

Fig. 18 Relationship between NDVI and NDWI for kharif months. The correlation is significant at a 0.01 level

availability, particularly during rabi, are essential. Additionally, the development of crop varieties resilient to water stress will benefit the region. While the present study was conducted at a monthly scale, finer temporal resolution (weekly or 15-day scale) would enable detection of short-duration droughts and improve monitoring accuracy. It is also important to note that broader climatic drivers such as ENSO, IOD, greenhouse gases, and aerosols can influence drought conditions in Vidarbha, and future studies should integrate these aspects for a deeper understanding and regional comparisons.

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Author contributions All authors contributed to the study conception and design. Amit G. Dhorde devised the main conceptual ideas and proof outline. Material preparation, data collection and analysis were performed by Anurag R. Marwade and Swapnil S. Vyas. Amit G. Dhorde and Anurag R. Marwade wrote the manuscript with input from all authors. In the last, all authors read and approved the final manuscript.

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Data availability All data generated or analyzed during this study are included in this article.

Declarations

Ethical approval This study does not contain any studies with human or animal subjects performed by any of the authors.

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Competing interests The authors declare no competing interests.

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